# Electronic Devices & Circuits Laboratory
# Experiment 2: BJT Characteristics

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# 1 Aim

- To plot the input and output characteristics of a BJT in the Common Emitter (CE) configuration, and to determine the current gain $\beta$, input resistance $r_i$, and output resistance $r_o$.
- To plot the input and output characteristics of a BJT in the Common Base (CB) configuration, and to determine the current gain $\alpha$, input resistance $r_i$, and output resistance $r_o$.
- To plot the input and output characteristics of a BJT in the Common Collector (CC) configuration, and to determine the current gain $\gamma$, input resistance $r_i$, and output resistance $r_o$.
- To verify the relationships $\beta = \alpha/(1-\alpha)$ and $\gamma = \beta + 1$ from the measured data.

# 2 Apparatus / Components

| S.No. | Component / Equipment | Specification | Quantity |
|---|---|---|---|
| 1 | BC547 transistor | NPN, TO-92 package | 1 |
| 2 | Regulated DC power supply | 0-30 V, dual output | 2 |
| 3 | Milliammeter | 0-10 mA and 0-30 mA range | 2 |
| 4 | Microammeter | 0-500 $\mu$A range | 1 |
| 5 | Voltmeter | 0-1 V and 0-30 V range | 2 |
| 6 | Fixed resistors | 1 k$\Omega$, 4.7 k$\Omega$, 10 k$\Omega$ | as needed |
| 7 | Breadboard, connecting wires, probes | Required | as needed |

Table 1: List of components required.

# 3 Common Emitter (CE) Configuration

## 3.1 Theory

A Bipolar Junction Transistor (BJT) is a three-terminal device - emitter (E), base (B), and collector (C) - formed by two back-to-back PN junctions. In normal (active-region) operation, the base-emitter junction is forward-biased and the base-collector junction is reverse-biased. A small base current controls a much larger collector current, giving current gain.

In the CE configuration, the emitter is common to both input (base-emitter) and output (collector-emitter) loops. The input characteristic ($I_B$ vs $V_{BE}$ at constant $V_{CE}$) resembles a diode's forward characteristic. The output characteristic ($I_C$ vs $V_{CE}$ at constant $I_B$) shows a steep rise in the saturation region followed by a nearly flat active region, whose slight upward slope reflects a finite output resistance. The current gain $\beta = \Delta I_C/\Delta I_B$ is typically 50-300.

## 3.2 Circuit Connections and Procedure

**Output Characteristics:**

1. Connect the transistor in CE configuration: emitter grounded (common), base current supplied through a 10 kΩ resistor from the base supply with a microammeter (0-500 $\mu$A) in series, and collector connected through a milliammeter (0-10 mA) to the collector supply.

2. Connect a voltmeter (0-1 V) across base-emitter and a voltmeter (0-30 V) across collector-emitter.

3. Set the base supply to give $I_B = 20\mu A$ (adjust and confirm on the microammeter) and keep it fixed for this run.

4. Starting with $V_{CE} = 0$ V, increase $V_{CE}$ in steps of 0.5 V up to 2 V, then in steps of 2 V up to 20 V, recording $I_C$ at each step.

5. Repeat for $I_B = 40, 60, 80$, and 100 $\mu$A, tabulating $I_C$ vs $V_{CE}$ for each. Switch off the supply.

**Input Characteristics:**

1. Retain the CE circuit. Set $V_{CE} = 1$ V and hold it fixed for this run.

2. Increase $V_{BE}$ in steps of 0.05 V from 0 to 0.4 V, then in steps of 0.02 V up to 0.7 V, recording $I_B$ at each step.

3. Repeat for $V_{CE} = 5$ V and $V_{CE} = 10$ V. Switch off the supply.

## 3.3 Circuit and Simulation

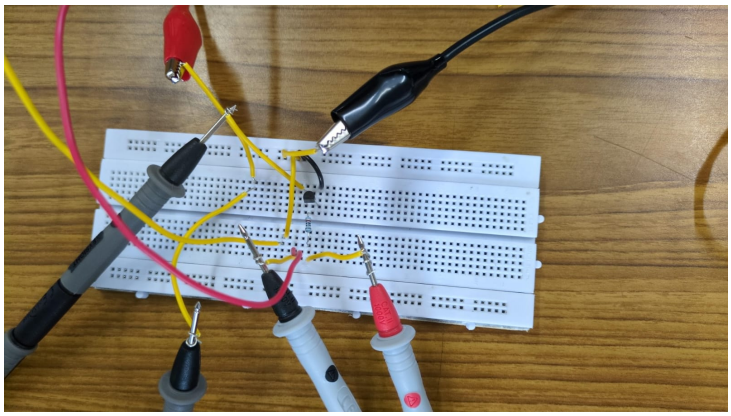


Figure 1: Hardware setup for CE configuration

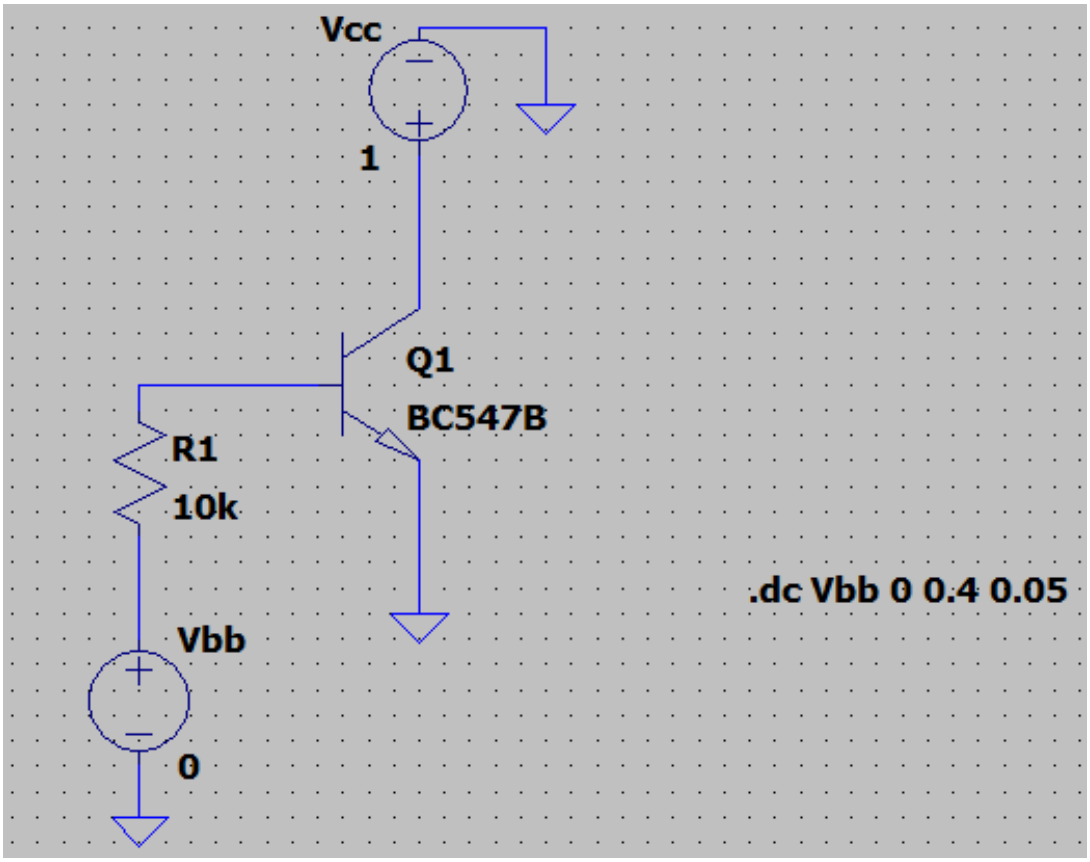



Figure 2: Simulation Circuit for CE Input Characteristics

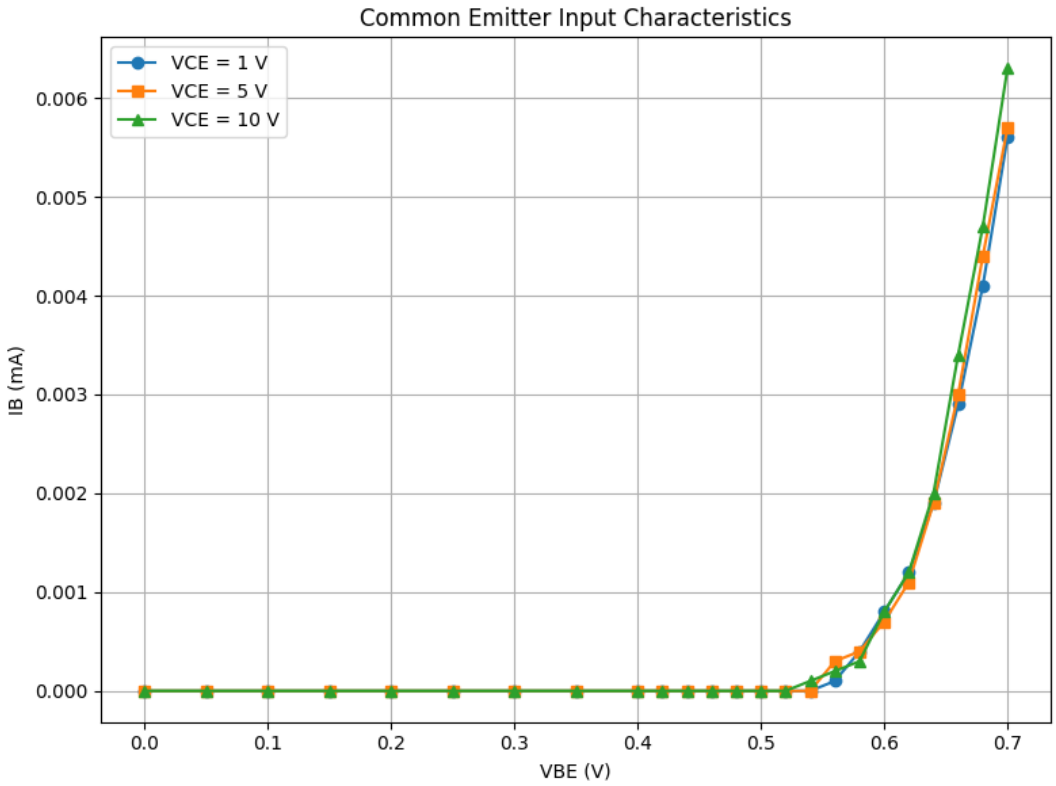



Figure 3: Simulated CE Input Characteristics ($I_B$ vs $V_{BE}$)

### 3.3.1 CE Input Simulations

The following simulations show the input characteristics at varying $V_{CE}$ values.

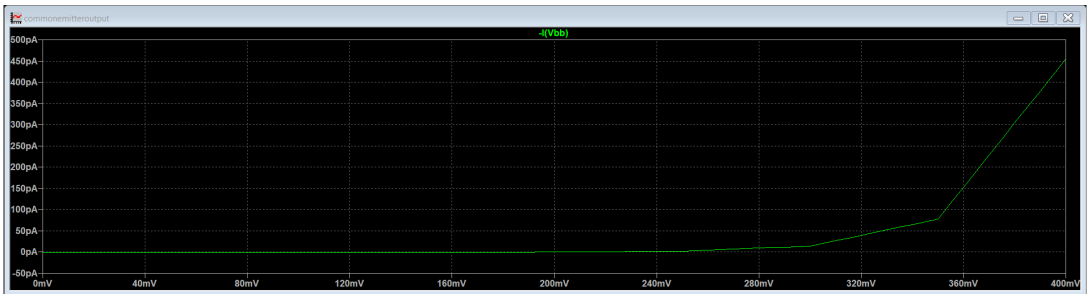



Figure 4: CE Input $V_{CE} = 1V$ (Low Range)

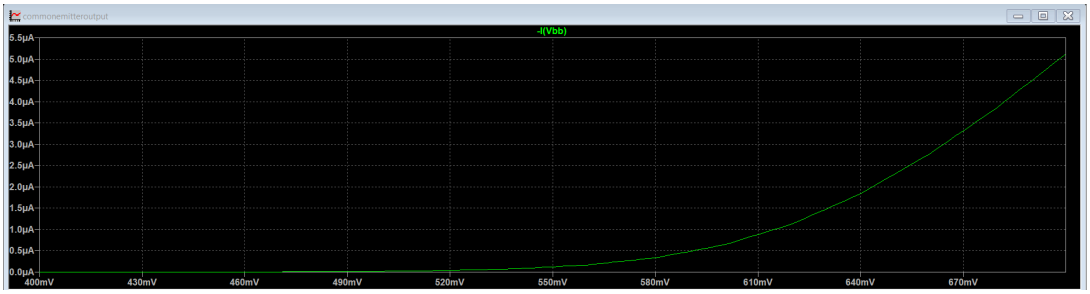



Figure 5: CE Input $V_{CE} = 1V$ (High Range)

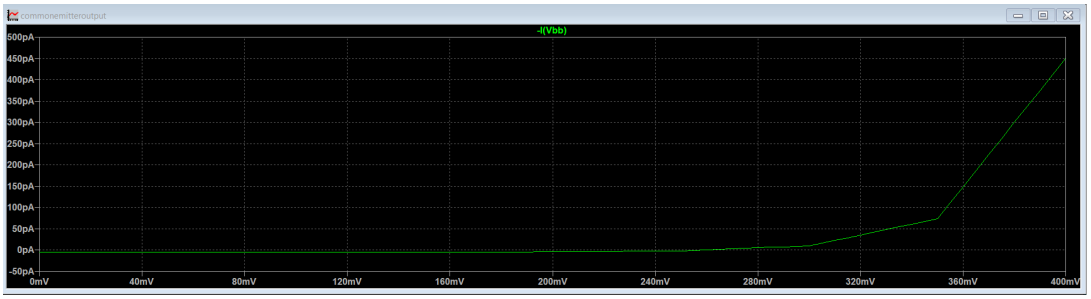



Figure 6: CE Input $V_{CE} = 5V$ (Low Range)

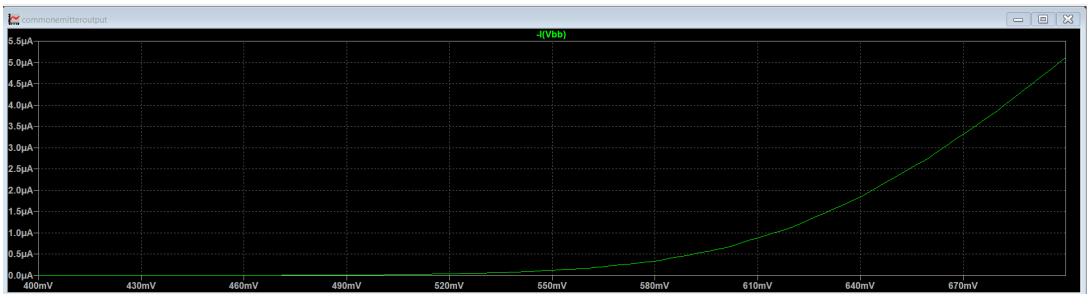



Figure 7: CE Input $V_{CE} = 5V$ (High Range)

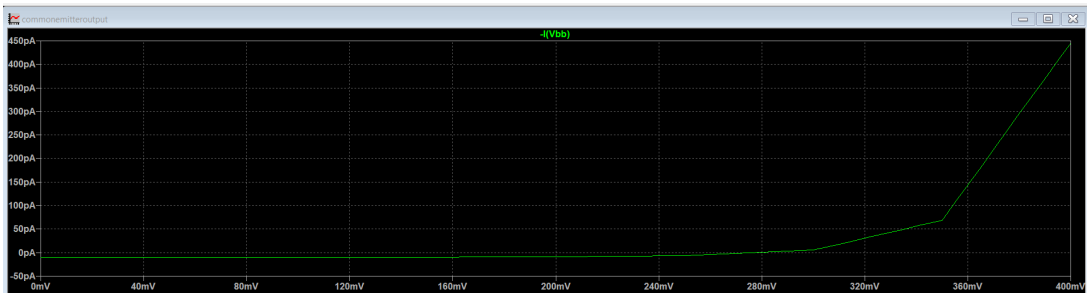



Figure 8: CE Input $V_{CE} = 10V$ (Low Range)

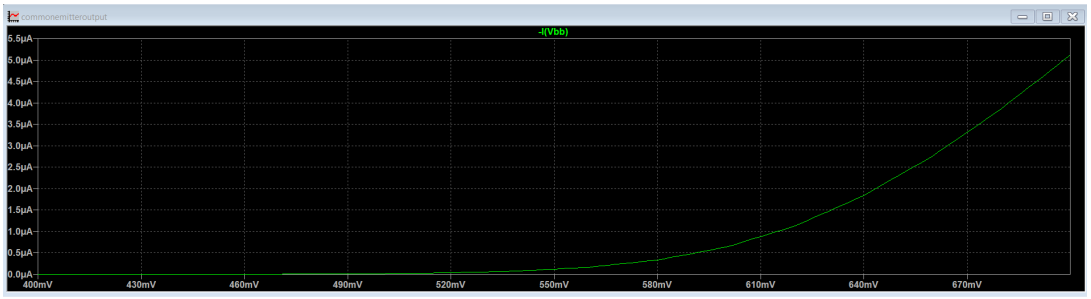



Figure 9: CE Input $V_{CE} = 10V$ (High Range)

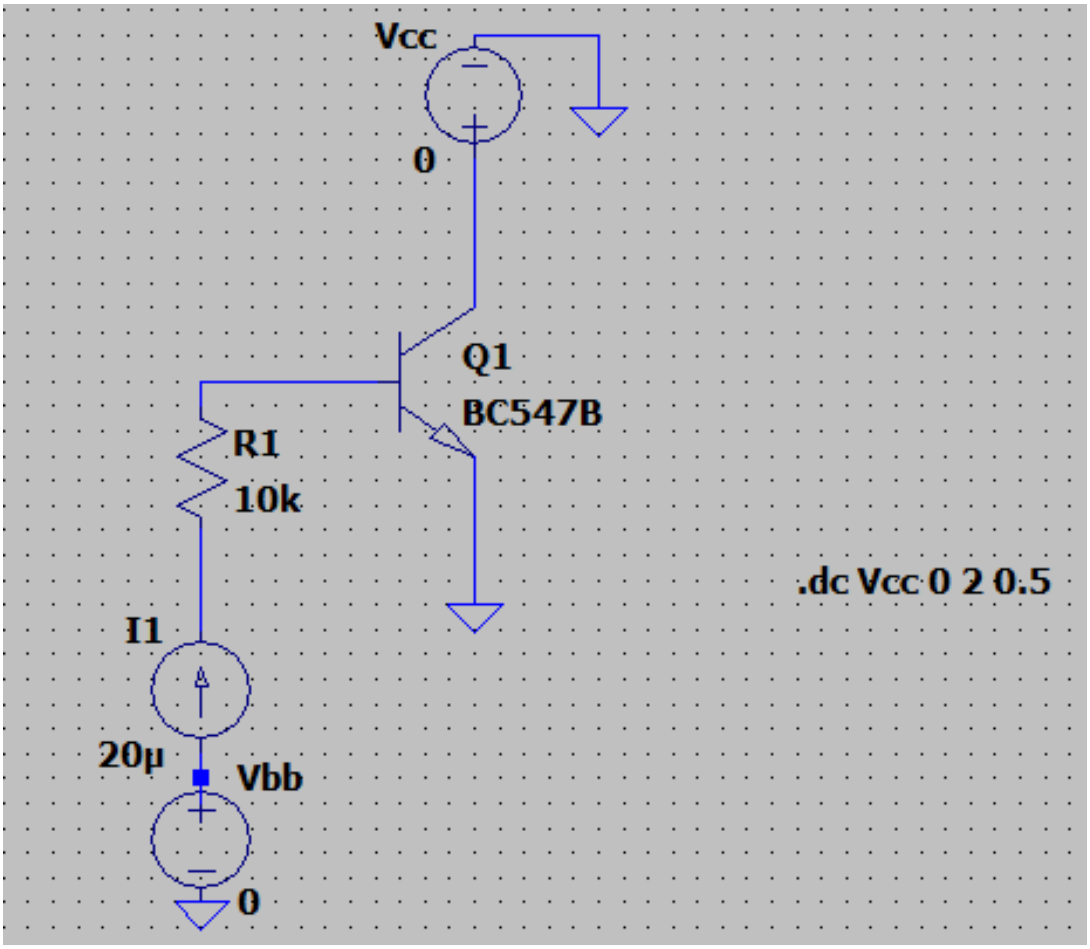



Figure 10: Simulation Circuit for CE Output Characteristics

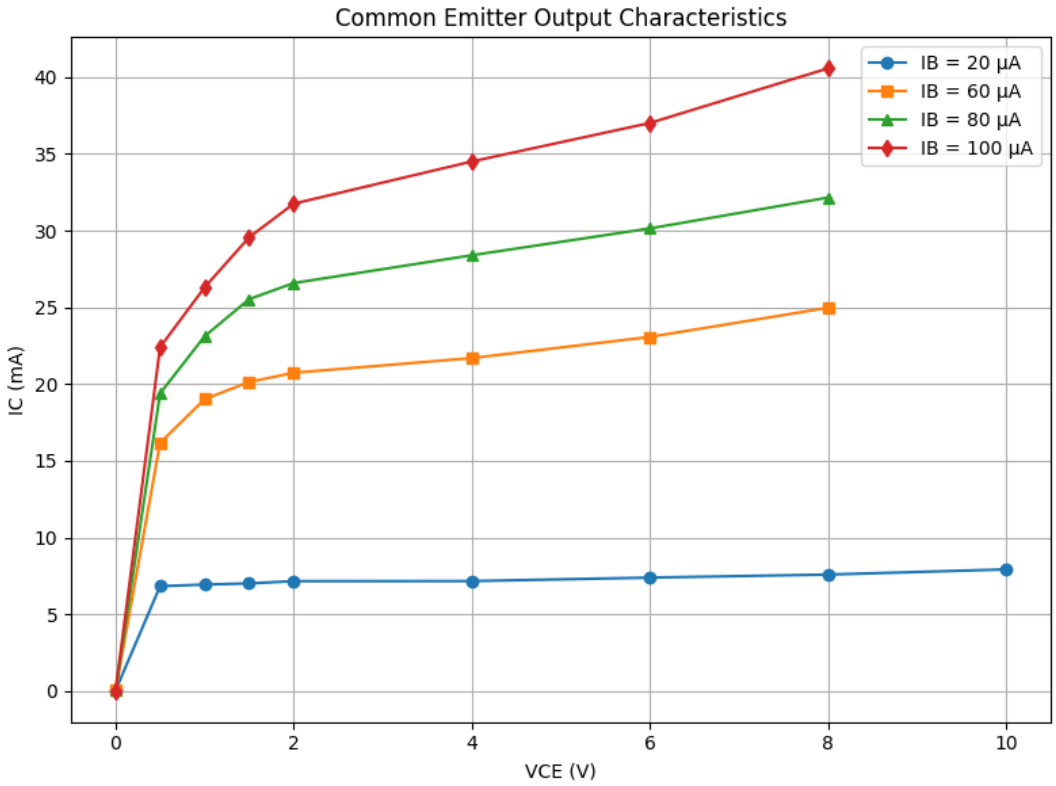



Figure 11: Simulated CE Output Characteristics ($I_C$ vs $V_{CE}$)

There is a slight increase in the active region in the $I_c$ value due to the increase in $V_{CE}$ which causes an reverse bias increase in the BC junction, which widens the depletion layer making the base thinner.

### 3.3.2 CE Output Simulations

The following images show the simulation results for varying base currents. The 'a' images represent the voltage sweep in steps of 0.5 V up to 2 V, and 'b' images represent steps of 2 V up to 20 V.

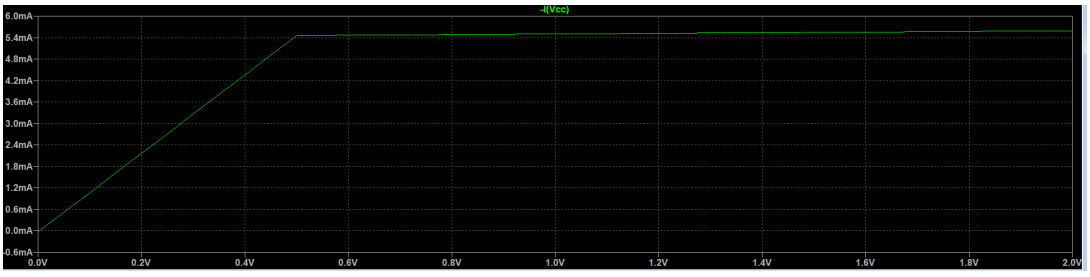



Figure 12: CE Output $I_B = 20\mu A$ (0 to 2V)

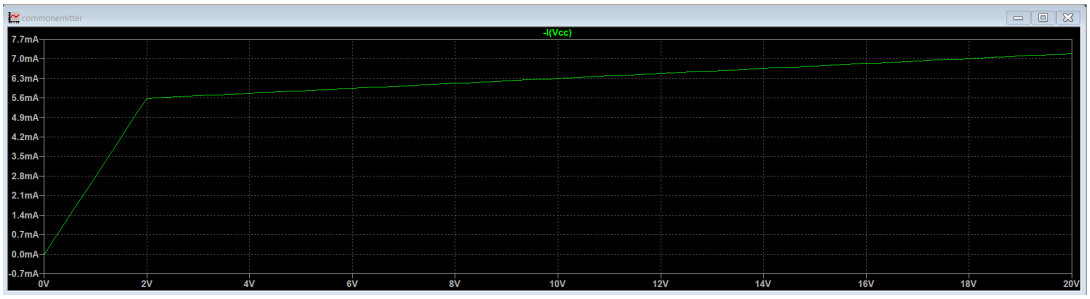



Figure 13: CE Output $I_B = 20\mu A$ (2 to 20V)

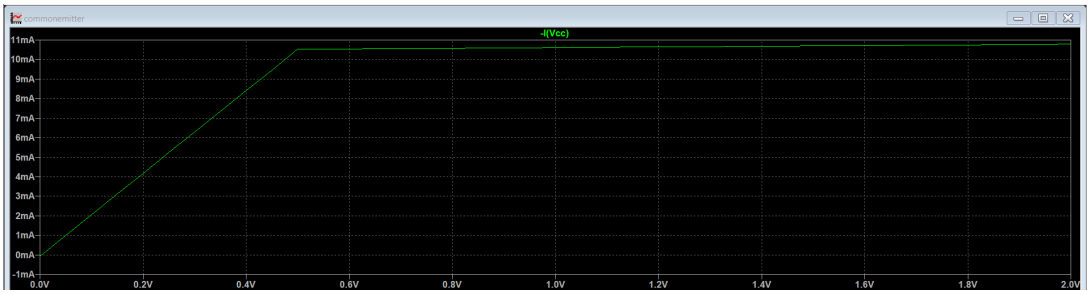



Figure 14: CE Output $I_B = 40\mu A$ (0 to 2V)

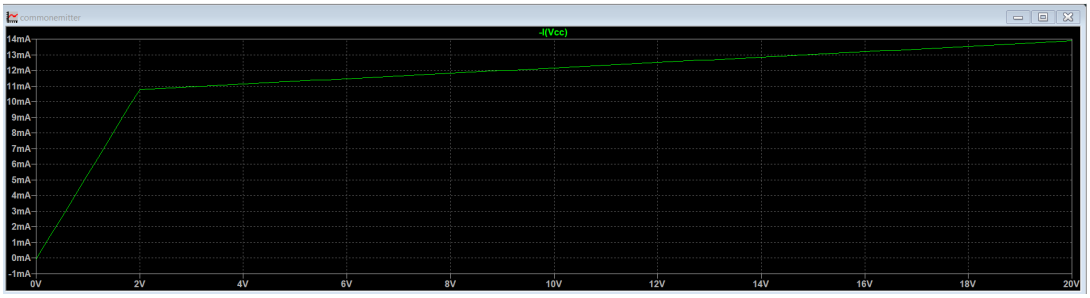



Figure 15: CE Output $I_B = 40\mu A$ (2 to 20V)

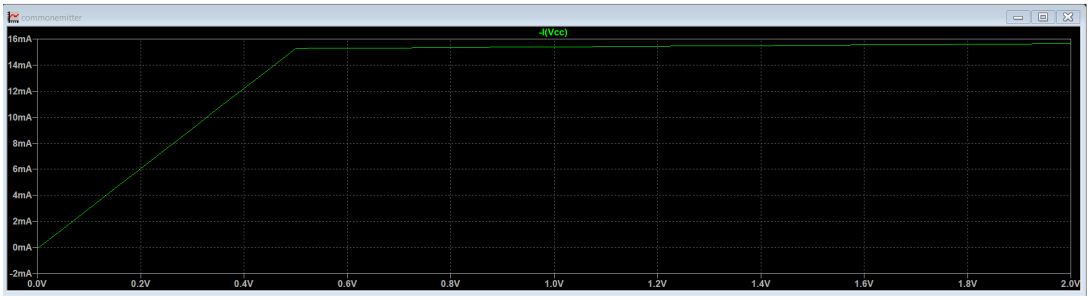



Figure 16: CE Output $I_B = 60\mu A$ (0 to 2V)

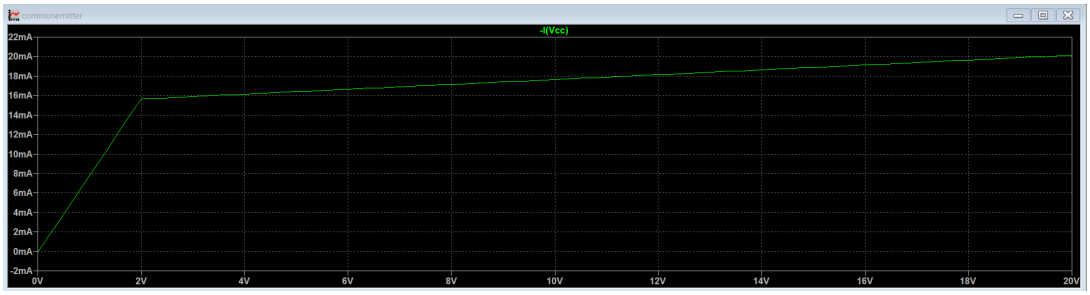



Figure 17: CE Output $I_B = 60\mu A$ (2 to 20V)

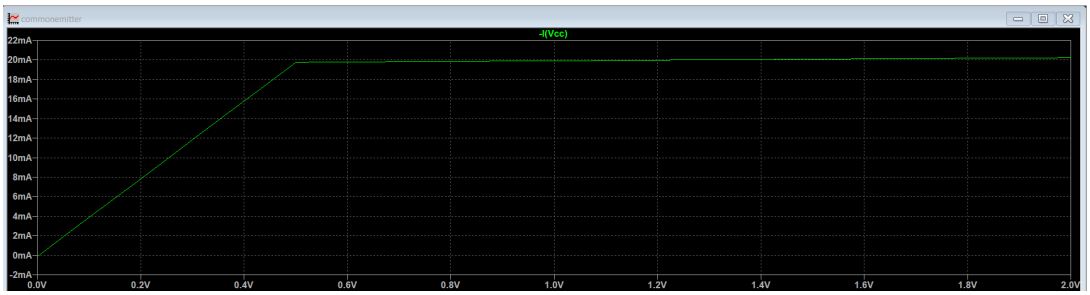



Figure 18: CE Output $I_B = 80\mu A$ (0 to 2V)

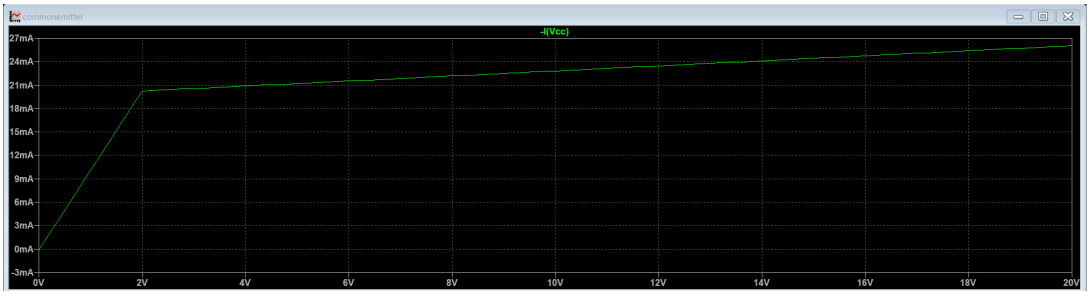



Figure 19: CE Output $I_B = 80\mu A$ (2 to 20V)

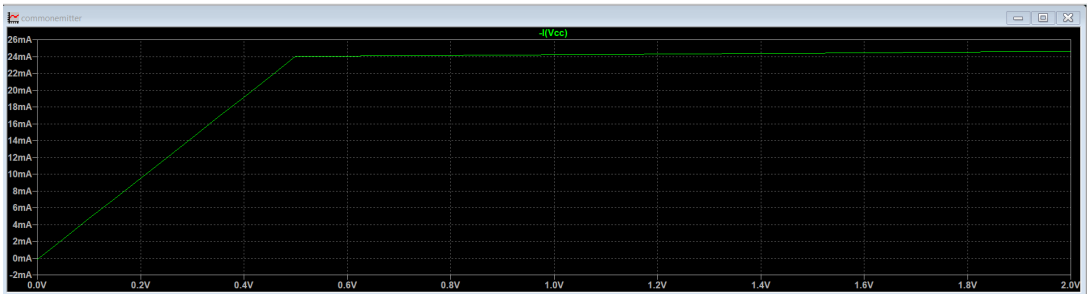



Figure 20: CE Output $I_B = 100\mu A$ (0 to 2V)

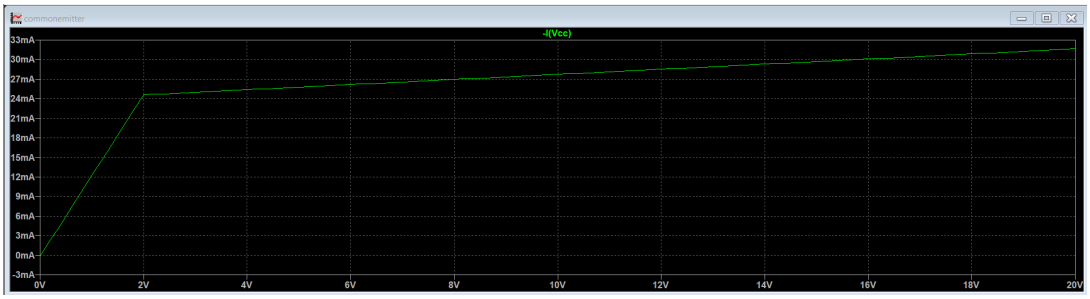



Figure 21: CE Output $I_B = 100\mu A$ (2 to 20V)

## 3.4 Observation Tables

Table 2: CE Output Characteristics

| **S.No.** | $V_{CE}$ **(V)** | $I_C$ **(mA) at** $I_B = 20\mu A$ | $I_C$ **(mA) at** $I_B = 60\mu A$ | $I_C$ **(mA) at** $I_B = 100\mu A$ |
|---|---|---|---|---|
| 1 | 0.0 | 0.031 | 0.031 | 0.0026 |
| 2 | 0.5 | 6.825 | 16.170 | 22.4000 |
| 3 | 1.0 | 6.930 | 19.015 | 26.3010 |
| 4 | 1.5 | 7.005 | 20.132 | 29.5600 |
| 5 | 2.0 | 7.155 | 20.732 | 31.7500 |
| 6 | 4.0 | 7.158 | 21.691 | 34.5000 |
| 7 | 6.0 | 7.385 | 23.070 | 37.0100 |
| 8 | 8.0 | 7.583 | 24.970 | 40.5700 |

Table 3: CE Input Characteristics

| **S.No.** | $V_{BE}$ **(V)** | $I_B$ **(mA) at** $V_{CE} = 1V$ | $I_B$ **(mA) at** $V_{CE} = 5V$ | $I_B$ **(mA) at** $V_{CE} = 10V$ |
|---|---|---|---|---|
| 1 | 0.00 | 0 | 0 | 0 |
| 2 | 0.05 | 0 | 0 | 0 |
| 3 | 0.10 | 0 | 0 | 0 |
| 4 | 0.15 | 0 | 0 | 0 |
| 5 | 0.20 | 0 | 0 | 0 |
| 6 | 0.25 | 0 | 0 | 0 |
| 7 | 0.30 | 0 | 0 | 0 |
| 8 | 0.35 | 0 | 0 | 0 |
| 9 | 0.40 | 0 | 0 | 0 |
| 10 | 0.42 | 0 | 0 | 0 |
| 11 | 0.44 | 0 | 0 | 0 |
| 12 | 0.46 | 0 | 0 | 0 |
| 13 | 0.48 | 0 | 0 | 0 |
| 14 | 0.50 | 0 | 0 | 0 |
| 15 | 0.52 | 0 | 0 | 0 |
| 16 | 0.54 | 0 | 0 | 0.0001 |
| 17 | 0.56 | 0.0001 | 0.0003 | 0.0002 |
| 18 | 0.58 | 0.0004 | 0.0004 | 0.0003 |
| 19 | 0.60 | 0.0008 | 0.0007 | 0.0008 |
| 20 | 0.62 | 0.0012 | 0.0011 | 0.0012 |
| 21 | 0.64 | 0.0019 | 0.0019 | 0.0020 |
| 22 | 0.66 | 0.0029 | 0.0030 | 0.0034 |
| 23 | 0.68 | 0.0041 | 0.0044 | 0.0047 |
| 24 | 0.70 | 0.0056 | 0.0057 | 0.0063 |

## 3.5 Calculations from Observation Data

**Current gain $\beta$:** Using the CE output table at $V_{CE} = 4.0$ V (active region, away from the saturation knee):

$$\beta_{20\to60\mu A} = \frac{\Delta I_C}{\Delta I_B} = \frac{21.691 - 7.158 \text{ mA}}{40\ \mu A} \approx 363$$

$$\beta_{60\to100\mu A} = \frac{34.500 - 21.691 \text{ mA}}{40\ \mu A} \approx 320$$

$$\beta_{20\to100\mu A} = \frac{34.500 - 7.158 \text{ mA}}{80\ \mu A} \approx 342$$

The measured $\beta$ is thus in the range **320–363**, consistent with the BC547 datasheet range (110–800 depending on the sample, typically 200–400).

**Output resistance $r_o$:** From the same table, taking $\Delta V_{CE}$ between 4.0 V and 8.0 V at fixed $I_B$:

- At $I_B = 20\ \mu A$: $r_o = \dfrac{8.0 - 4.0 \text{ V}}{(7.583 - 7.158) \text{ mA}} \approx 9.4 \text{ k}\Omega$
- At $I_B = 60\ \mu A$: $r_o = \dfrac{4.0 \text{ V}}{(24.970 - 21.691) \text{ mA}} \approx 1.22 \text{ k}\Omega$

- At $I_B = 100\ \mu A$: $r_o = \dfrac{4.0\text{ V}}{(40.570 - 34.500)\text{ mA}} \approx 0.66\text{ k}\Omega$

$r_o$ decreases as $I_B$ (and hence $I_C$) increases, as expected from the Early-effect slope of the output family.

**Input resistance $r_i$:** Using the CE input table, in the exponential (forward-conducting) region between $V_{BE} = 0.64$ V and 0.70 V:

$$r_{i,\,V_{CE}=1V} = \frac{\Delta V_{BE}}{\Delta I_B} = \frac{0.70 - 0.68\text{ V}}{(0.0056 - 0.0041)\text{ mA}} \approx 13.3\text{ k}\Omega$$

$$r_{i,\,V_{CE}=5V} = \frac{0.02\text{ V}}{(0.0057 - 0.0044)\text{ mA}} \approx 15.4\text{ k}\Omega$$

$$r_{i,\,V_{CE}=10V} = \frac{0.02\text{ V}}{(0.0063 - 0.0047)\text{ mA}} \approx 12.5\text{ k}\Omega$$

Taking the step before that ($V_{BE} = 0.64 \rightarrow 0.66$ V) gives $r_i$ in the range 14–20 kΩ across the three curves, so overall $r_i \approx$ 13–20 kΩ near cut-in, decreasing towards the lower end of that range as $V_{BE}$ increases further (consistent with the diode-like $I_B$–$V_{BE}$ curve becoming steeper, i.e. lower dynamic resistance, at higher forward bias). This is a moderate input resistance, as expected for CE — higher than CB ($\approx 830\ \Omega$,) but much lower than CC ($\approx 8.3$ MΩ), matching the expected ordering $r_{i,CB} < r_{i,CE} < r_{i,CC}$.

# 4 Common Base (CB) Configuration

## 4.1 Theory

In the CB configuration, the base is common to both loops. The input characteristic ($I_E$ vs $V_{EB}$ at constant $V_{CB}$) is again diode-like. The output characteristic ($I_C$ vs $V_{CB}$ at constant $I_E$) is extremely flat, since the collector current is set almost entirely by the emitter current and is nearly independent of $V_{CB}$ giving CB circuits a very high output resistance. The current gain $\alpha = \Delta I_C / \Delta I_E$ is always slightly less than 1 (typically 0.95-0.99).

## 4.2 Circuit Connections and Procedure

**Output Characteristics:**

1. Connect the transistor in CB configuration: base grounded (common), emitter current supplied through a 1 kΩ resistor from the emitter supply with a milliammeter (0-10 mA) in series, and collector connected through a milliammeter (0-10 mA) to the collector supply.

2. Connect a voltmeter (0-1 V) across emitter-base and a voltmeter (0-30 V) across collector-base.

3. Set the emitter supply to give $I_E = 2$ mA and hold it fixed for this run.

4. Starting with $V_{CB} = 0$ V, increase $V_{CB}$ in steps of 0.5 V up to 2 V, then in steps of 2 V up to 20 V, recording $I_C$ at each step.

5. Repeat for $I_E = 4, 6, 8$, and 10 mA. Switch off the supply.

**Input Characteristics:**

1. Retain the CB circuit. Set $V_{CB} = 0$ V and hold it fixed for this run.
2. Increase $V_{EB}$ in steps of 0.05 V from 0 to 0.4 V, then in steps of 0.02 V up to 0.7 V, recording $I_E$ at each step.
3. Repeat for $V_{CB} = 5$ V and $V_{CB} = 10$ V. Switch off the supply.

## 4.3 Circuit and Simulation

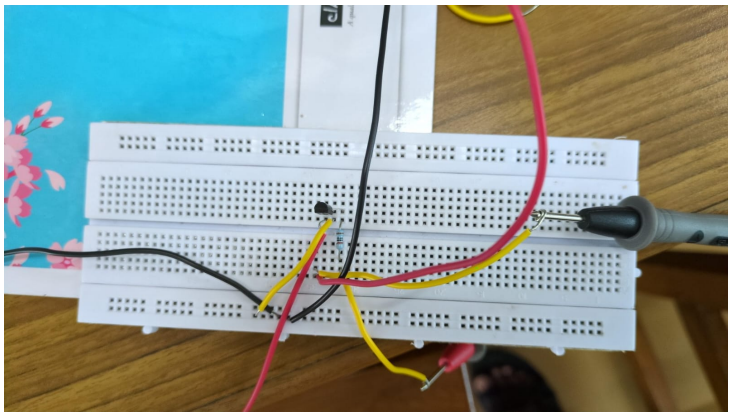


Figure 22: Hardware setup for CB configuration

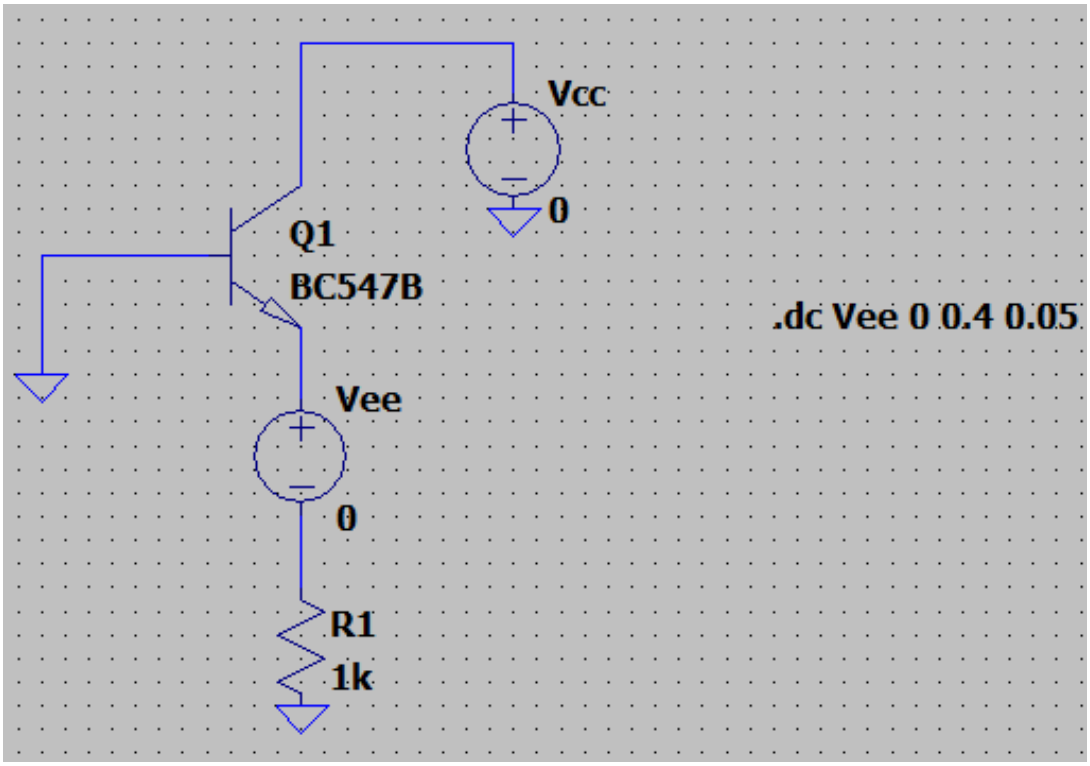



Figure 23: Simulation Circuit for CB Configuration

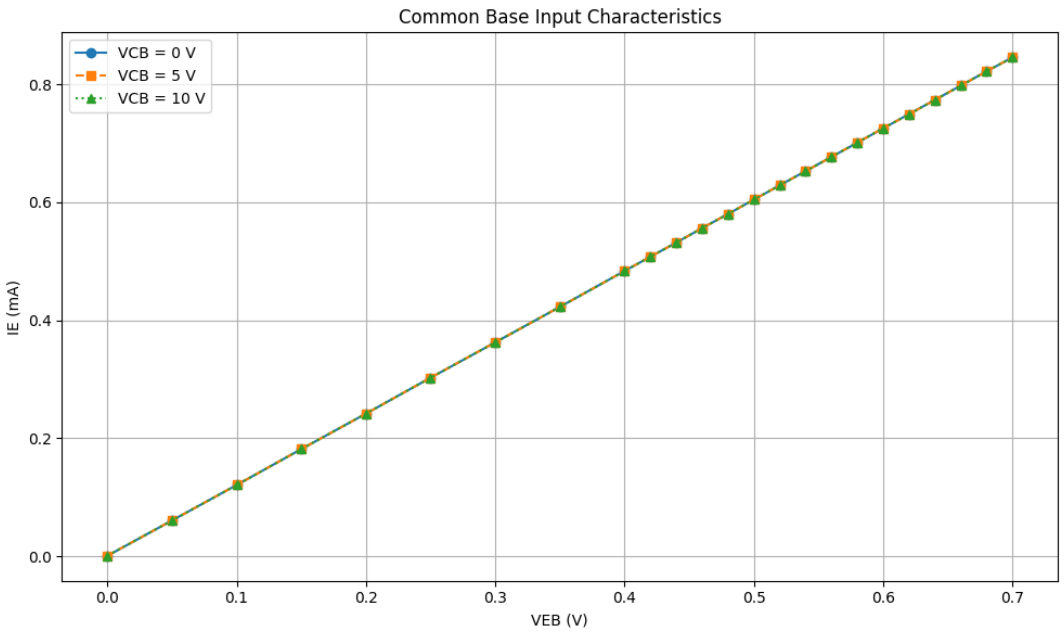



Figure 24: Simulated CB Input Characteristics ($I_E$ vs $V_{EB}$)

### 4.3.1 CB Input Simulations

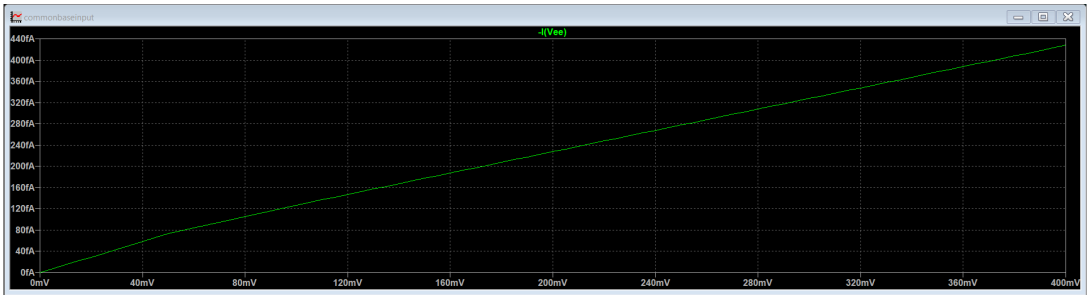



Figure 25: CB Input $V_{CB} = 0V$ (Low Range)

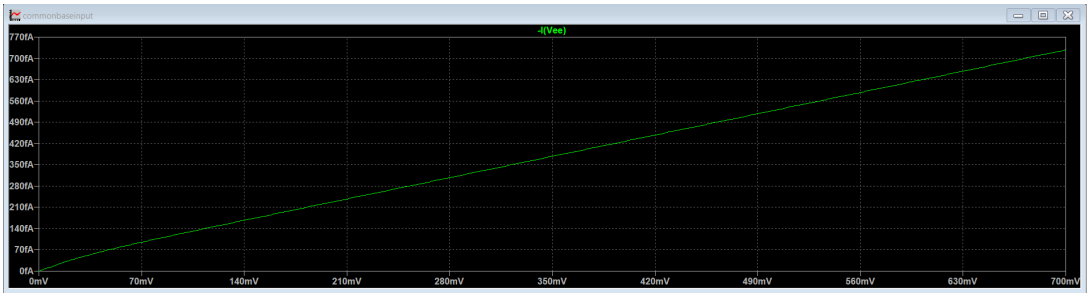



Figure 26: CB Input $V_{CB} = 0V$ (High Range)

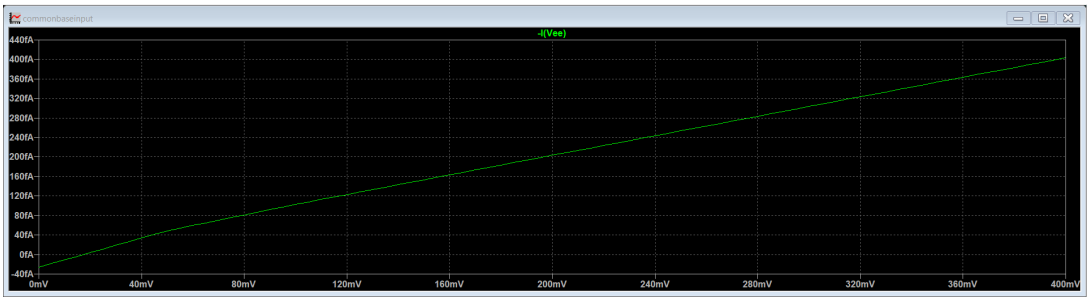



Figure 27: CB Input $V_{CB} = 5V$ (Low Range)

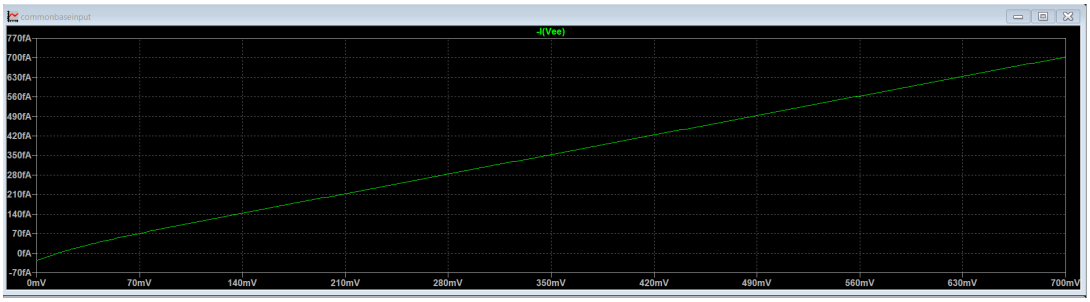



Figure 28: CB Input $V_{CB} = 5V$ (High Range)

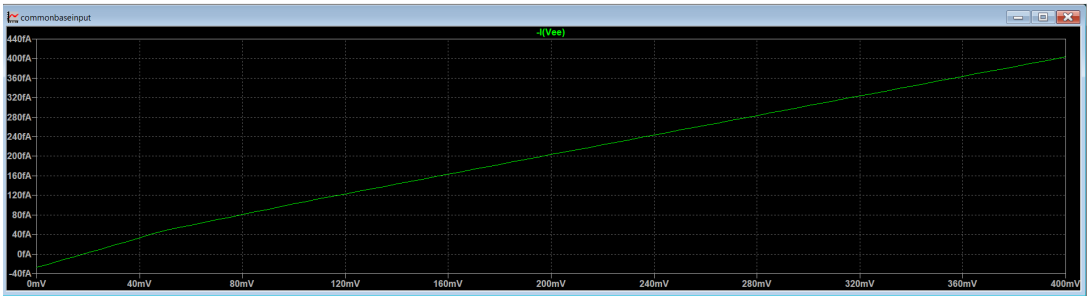



Figure 29: CB Input $V_{CB} = 10V$ (Low Range)

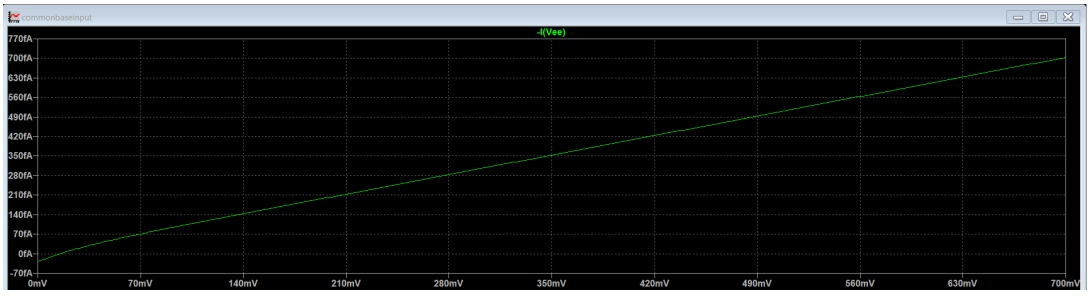



Figure 30: CB Input $V_{CB} = 10V$ (High Range)

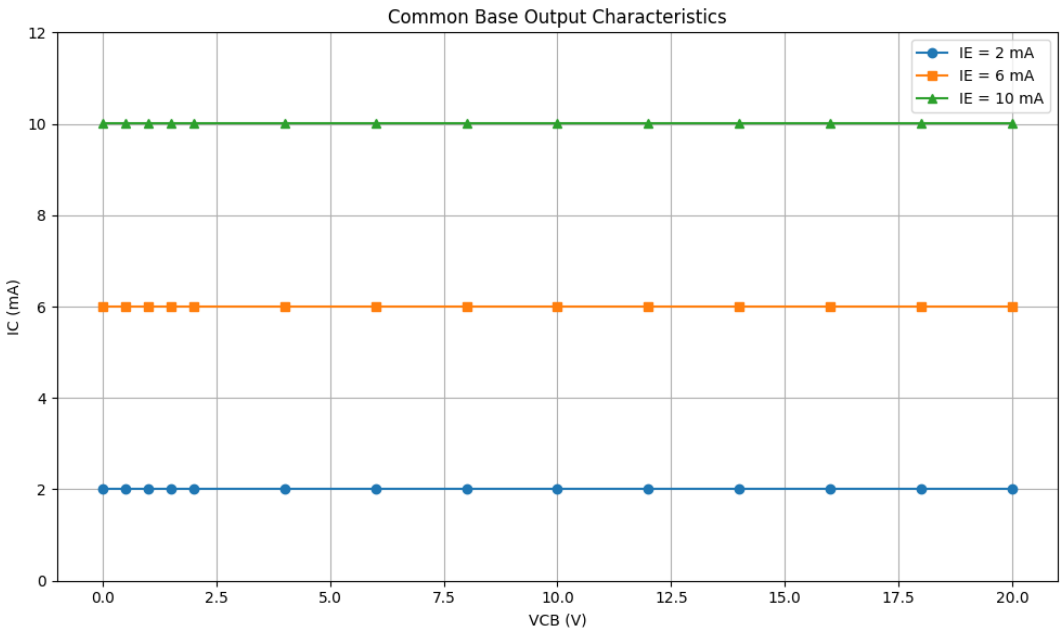



Figure 31: Simulated CB Output Characteristics ($I_C$ vs $V_{CB}$)

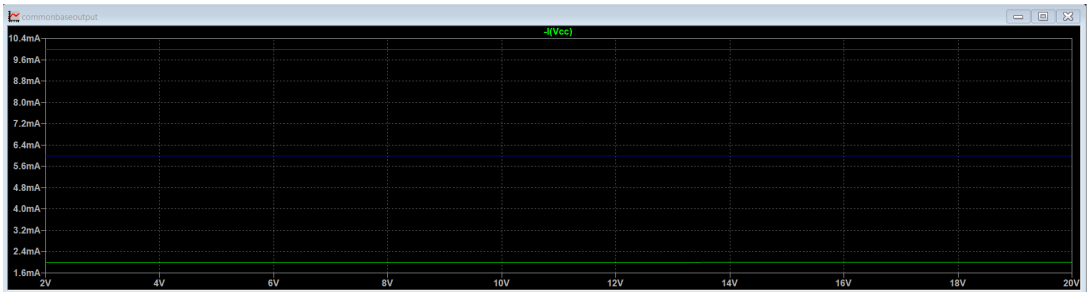



Figure 32: Simulated CB Output Characteristics Detailed Simulation

## 4.4 Observation Tables

Table 4: CB Output Characteristics

| **S.No.** | $V_{CB}$ **(V)** | $I_C$ **(mA) at** $I_E = 2$ **mA** | $I_C$ **(mA) at** $I_E = 6$ **mA** | $I_C$ **(mA) at** $I_E = 10$ **mA** |
|---|---|---|---|---|
| 1 | 0.0 | 1.99 | 5.98 | 9.96 |
| 2 | 0.5 | 1.99 | 5.98 | 9.96 |
| 3 | 1.0 | 1.99 | 5.98 | 9.96 |
| 4 | 1.5 | 1.99 | 5.98 | 9.97 |
| 5 | 2.0 | 2.00 | 5.98 | 9.97 |
| 6 | 4.0 | 2.00 | 5.98 | 9.98 |
| 7 | 6.0 | 2.00 | 5.99 | 9.98 |
| 8 | 8.0 | 2.00 | 5.99 | 9.98 |

Table 5: CB Input Characteristics

| **S.No.** | $V_{EB}$ **(V)** | $I_E$ **(mA) at** $V_{CB} = 0V$ | $I_E$ **(mA) at** $V_{CB} = 5V$ | $I_E$ **(mA) at** $V_{CB} = 10V$ |
|---|---|---|---|---|
| 1 | 0.00 | 0.0007 | 0.0008 | 0.0008 |
| 2 | 0.05 | 0.0606 | 0.0606 | 0.0606 |
| 3 | 0.10 | 0.1207 | 0.1207 | 0.1207 |
| 4 | 0.15 | 0.1816 | 0.1816 | 0.1816 |
| 5 | 0.20 | 0.2417 | 0.2417 | 0.2417 |
| 6 | 0.25 | 0.3026 | 0.3026 | 0.3026 |
| 7 | 0.30 | 0.3625 | 0.3625 | 0.3625 |
| 8 | 0.35 | 0.4227 | 0.4227 | 0.4227 |

## 4.5 Calculations from Observation Data

**Current gain $\alpha$:** Using the CB output table,

$$\alpha = \frac{\Delta I_C}{\Delta I_E} = \frac{9.96 - 1.99 \text{ mA}}{10 - 2 \text{ mA}} = 0.99625$$

$$\beta = \frac{0.99625}{1 - 0.99625} \approx 266$$

**Output resistance $r_o$:** Using the re-measured CB output table, taking the full swept range $V_{CB} = 0$ V to 8.0 V at each fixed $I_E$:

$$r_{o,\, I_E = 2mA} = \frac{\Delta V_{CB}}{\Delta I_C} = \frac{8.0 \text{ V}}{(2.00 - 1.99) \text{ mA}} \approx 800 \text{ k}\Omega$$

$$r_{o,\, I_E = 6mA} = \frac{8.0 \text{ V}}{(5.99 - 5.98) \text{ mA}} \approx 800 \text{ k}\Omega$$

$$r_{o,\, I_E = 10mA} = \frac{8.0 \text{ V}}{(9.98 - 9.96) \text{ mA}} \approx 400 \text{ k}\Omega$$

So $r_o$ is in the range **400–800 k**$\Omega$, roughly two orders of magnitude higher than the CE-measured $r_o$ (0.66–9.4 k$\Omega$) and even higher than the CC-measured $r_o$ (2.8–7.5 k$\Omega$). This is consistent with CB theory: since $I_C$ is set almost entirely by $I_E$ and depends only very weakly on $V_{CB}$, the CB output characteristic is the flattest of the three configurations, giving by far the highest output resistance.

**Input resistance $r_i$:** Using the CB input table, in the forward-conducting region between $V_{EB} = 0.30$ V and 0.35 V:

$$r_i = \frac{\Delta V_{EB}}{\Delta I_E} = \frac{0.35 - 0.30 \text{ V}}{(0.4227 - 0.3625) \text{ mA}} \approx 831 \ \Omega$$

**Cross-check of $\beta = \alpha/(1 - \alpha)$:** The CE-measured $\beta$ (320–363) and the CB-derived $\beta$ from $\alpha = 0.99625$ ($\approx 266$) agree to within the same order of magnitude, which is a reasonable cross-check given how sensitive $\beta = \alpha/(1 - \alpha)$ is to small errors in $\alpha$ near 1.

# 5 Common Collector (CC) Configuration

## 5.1 Theory

In the CC configuration (also called an emitter follower), the collector is common to both loops. The output characteristic ($I_E$ vs $V_{EC}$ at constant $I_B$) closely resembles the CE output characteristic, since $I_E \approx I_C$. The input characteristic is normally plotted as $I_B$ vs $V_{BC}$ at constant $V_{EC}$, and falls off as the base-collector reverse bias increases. The current gain $\gamma = \Delta I_E / \Delta I_B \approx \beta + 1$, so CC circuits provide current gain close to that of the CE configuration but are used mainly for their high input resistance and low output resistance (voltage buffering).

## 5.2 Circuit Connections and Procedure

**Output Characteristics:**

1. Connect the transistor in CC configuration: collector grounded (common), base current supplied through a 10 k$\Omega$ resistor from the base supply with a microammeter (0-500 $\mu$A) in series, and emitter connected through a milliammeter (0-10 mA) to the emitter supply.

2. Connect a voltmeter (0-30 V) across base-collector and a voltmeter (0-30 V) across emitter-collector.

3. Set the base supply to give $I_B = 20\mu A$ and hold it fixed for this run.

4. Starting with $V_{EC} = 0$ V increase $V_{EC}$ in steps of 0.5 V up to 2 V, then in steps of 2 V up to 20 V, recording $I_E$ at each step.

5. Repeat for $I_B = 40, 60, 80$, and 100 $\mu$A. Switch off the supply.

**Input Characteristics:**

1. Retain the CC circuit. Set $V_{EC} = 1$ V and hold it fixed for this run.

2. Increase $V_{BC}$ in steps of 0.5 V from 0 to 10 V, recording $I_B$ at each step.

3. Repeat for $V_{EC} = 5$ V and $V_{EC} = 10$ V. Switch off the supply.

## 5.3 Circuit and Simulation

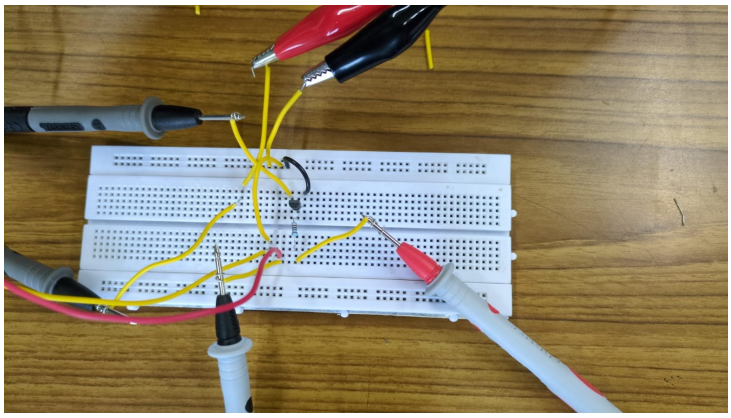


Figure 33: Hardware setup for CC configuration

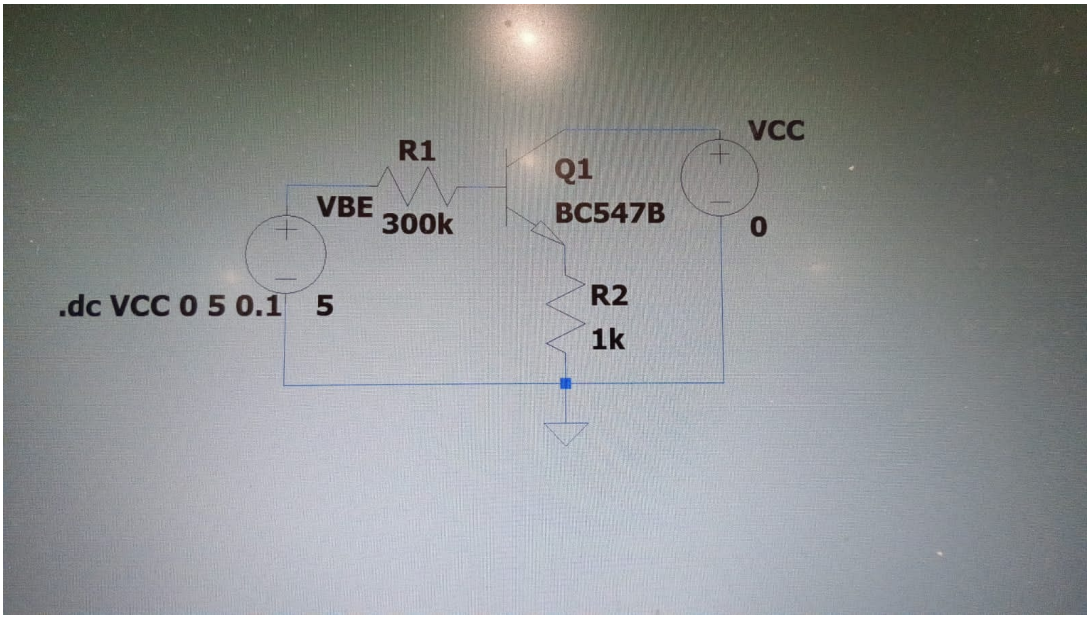



Figure 34: Simulation Circuit for CC Configuration

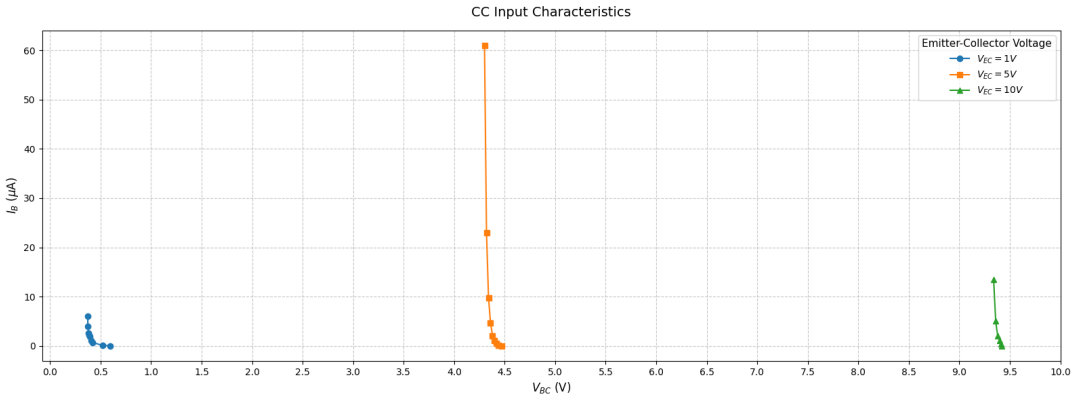



Figure 35: Experimental CC Input Characteristics($I_B$ vs $V_{BC}$)

### 5.3.1 CC Input Simulations

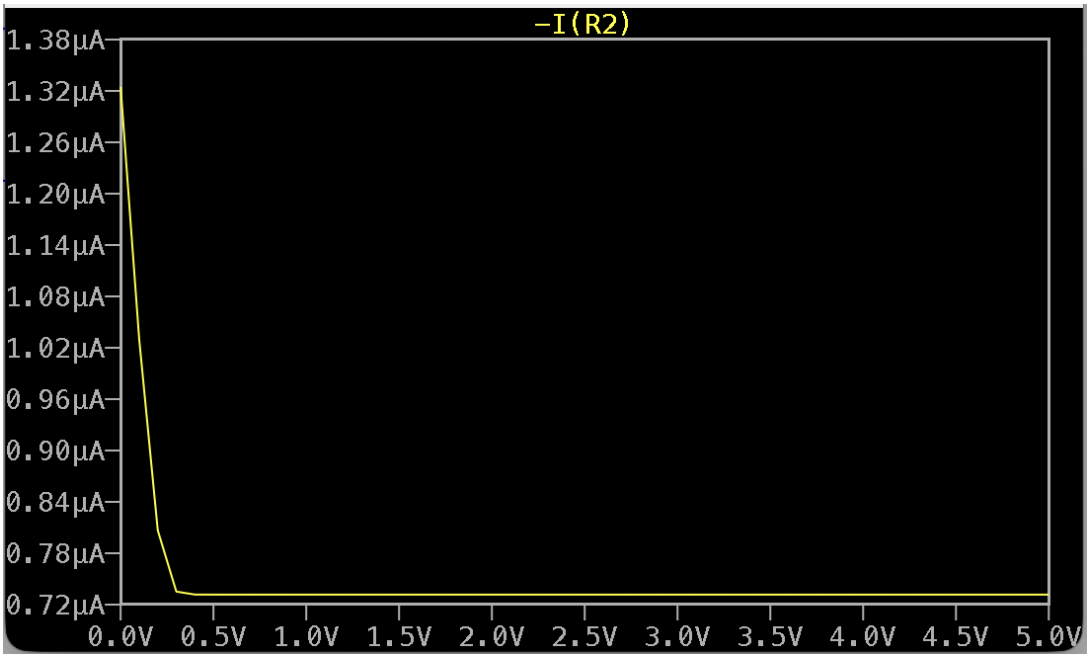



Figure 36: CC Input $V_{EC} = 1V$

Table 6: CC Input Characteristics

| **S.No.** | $V_{BC}(V)$ **for** $V_{EC} = 1V$ | $I_B$ **(**$\mu$**A) at** $V_{EC} = 1V$ | $V_{BC}(V)$ **for** $V_{EC} = 5V$ | $I_B$ **(**$\mu$**A) at** $V_{EC} = 5V$ | $V_{BC}(V)$ **for** $V_{EC} = 10V$ | $I_B$ **(**$\mu$**A) at** $V_{EC} = 10V$ |
|---|---|---|---|---|---|---|
| 1 | 0.593 | 0 | 4.47 | 0 | 9.42 | 0.0000 |
| 2 | 0.518 | 0.1 | 4.44 | 0.1 | 9.41 | 0.4 |
| 3 | 0.418 | 0.7 | 4.42 | 0.5 | 9.40 | 1 |
| 4 | 0.407 | 1 | 4.40 | 1 | 9.38 | 2.0 |
| 5 | 0.39 | 1.9 | 4.38 | 2 | 9.36 | 5.0 |
| 6 | 0.385 | 2.2 | 4.36 | 4.6 | 9.34 | 13.4 |
| 7 | 0.382 | 2.6 | 4.34 | 9.7 | | |
| 8 | 0.373 | 3.9 | 4.32 | 23 | | |
| 9 | 0.374 | 6 | 4.30 | 61 | | |

### 5.3.2 CC Output Simulations

The following images show the simulation results for varying base currents. The 'a' images represent the voltage sweep in steps of 0.5 V up to 2 V, and 'b' images represent steps of 2 V up to 20 V. (In the figures below, -I vs V is calculated), **i.e., SIMULATION GRAPHS ARE REFLECTED OVER THE X-AXIS**

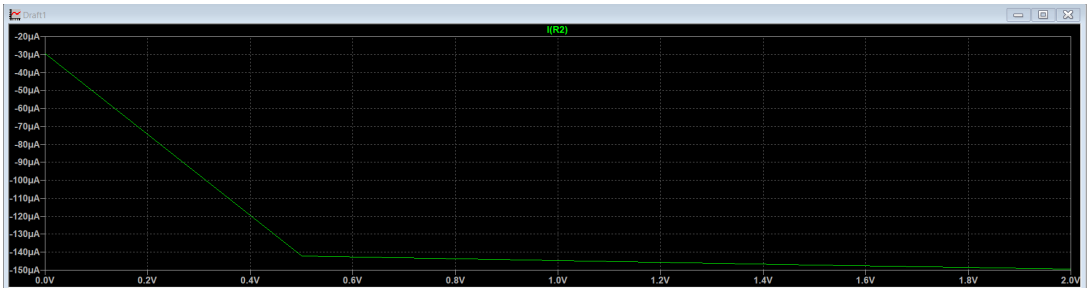



Figure 37: CC Output $I_B = 20\mu A$ (0 to 2V)

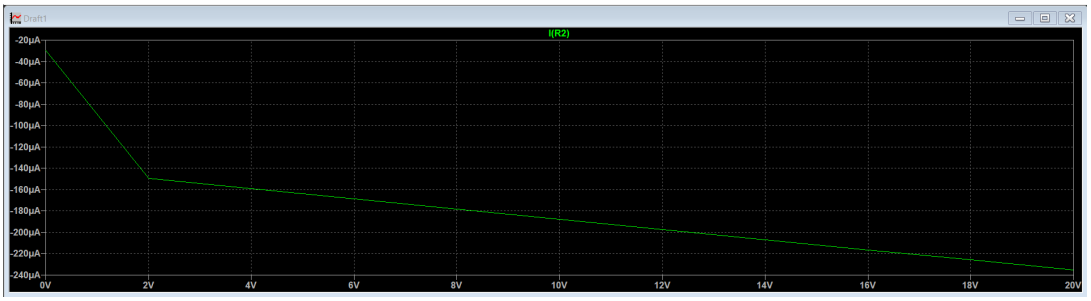



Figure 38: CC Output $I_B = 20\mu A$ (2 to 20V)

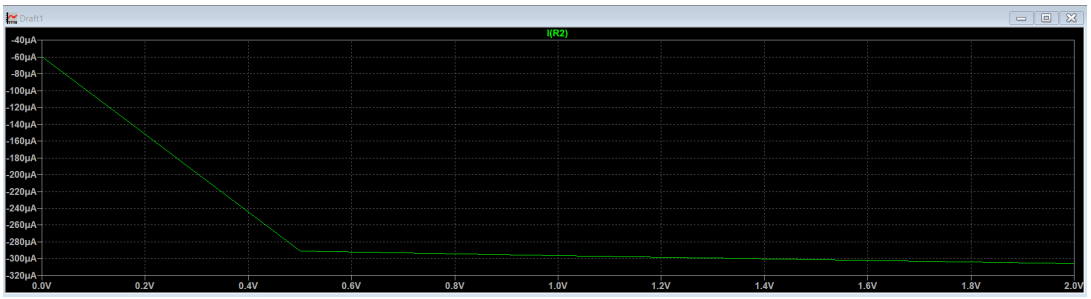



Figure 39: CC Output $I_B = 40\mu A$ (0 to 2V)

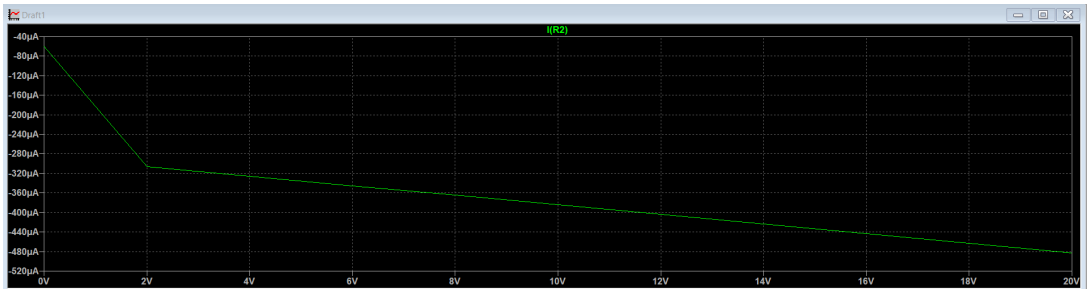



Figure 40: CC Output $I_B = 40\mu A$ (2 to 20V)

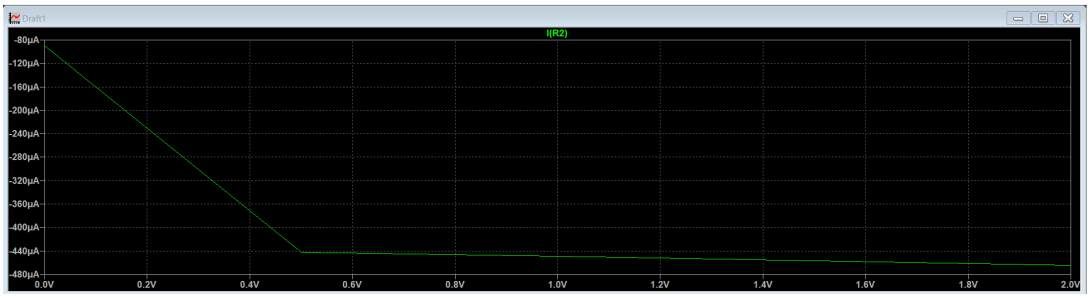



Figure 41: CC Output $I_B = 60\mu A$ (0 to 2V)

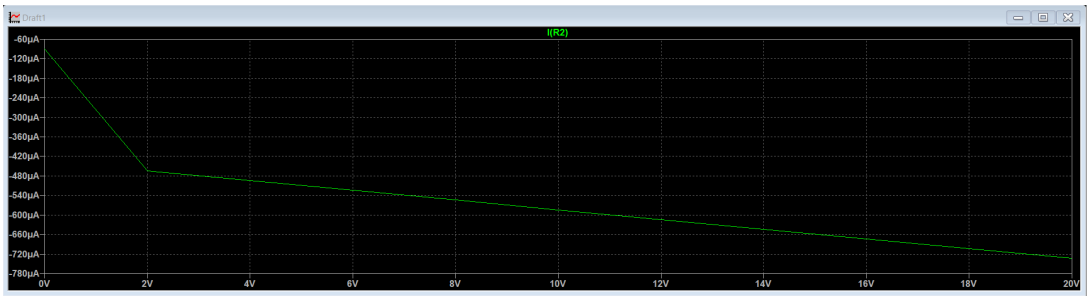



Figure 42: CC Output $I_B = 60\mu A$ (2 to 20V)

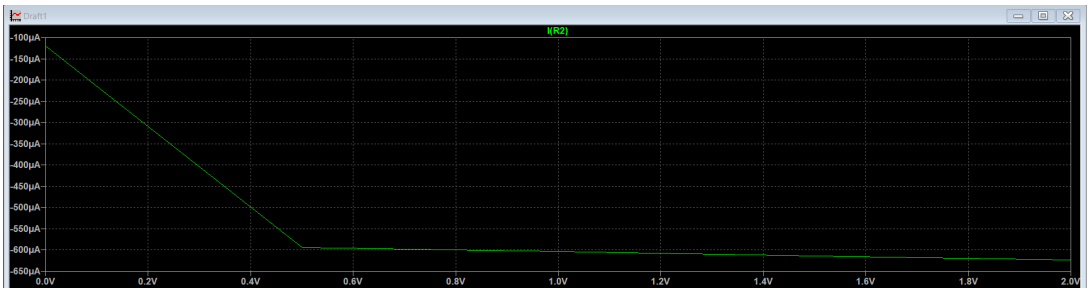



Figure 43: CC Output $I_B = 80\mu A$ (0 to 2V)

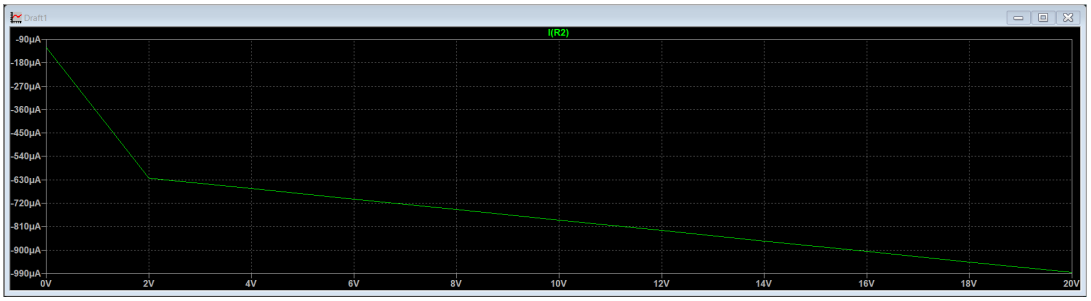



Figure 44: CC Output $I_B = 80\mu A$ (2 to 20V)

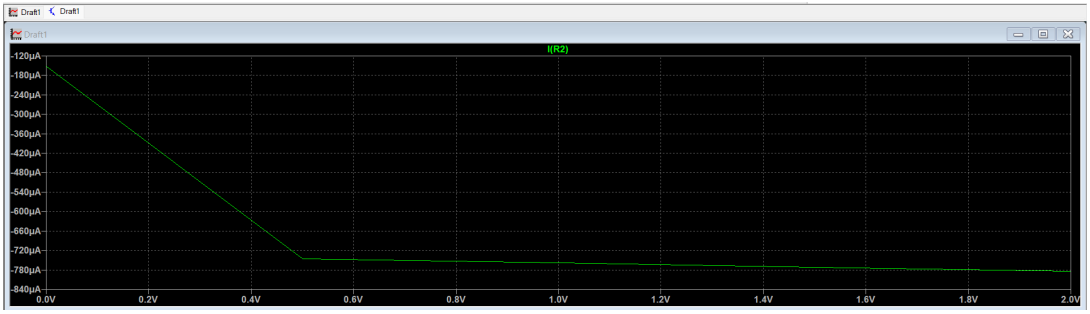



Figure 45: CC Output $I_B = 100\mu A$ (0 to 2V)

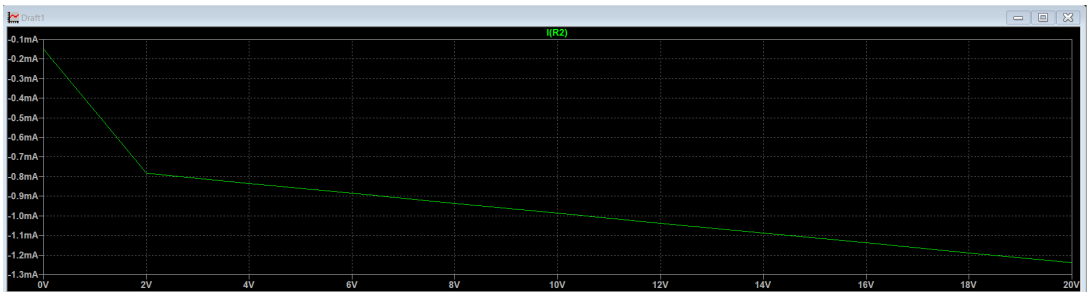



Figure 46: CC Output $I_B = 100\mu A$ (2 to 20V)

## 5.4 Observation Tables

Table 7: CC Output Characteristics

| **S.No.** | $V_{EC}$ **(V)** | $I_E$ **(mA) at** $I_B = 20\mu A$ | $I_E$ **(mA) at** $I_B = 60\mu A$ | $I_E$ **(mA) at** $I_B = 100\mu A$ |
|---|---|---|---|---|
| 1 | 0.0 | 0.0016 | 0.05 | 0.09 |
| 2 | 0.5 | 6.21 | 16.01 | 21.50 |
| 3 | 1.0 | 6.24 | 18.30 | 24.62 |
| 4 | 1.5 | 6.3 | 18.60 | 26.55 |
| 5 | 2.0 | 6.35 | 18.90 | 27.35 |
| 6 | 4.0 | 6.47 | 19.65 | 28.81 |
| 7 | 6.0 | 6.65 | 21.09 | 30.4 |
| 8 | 8.0 | 6.78 | 21.9 | 31.9 |

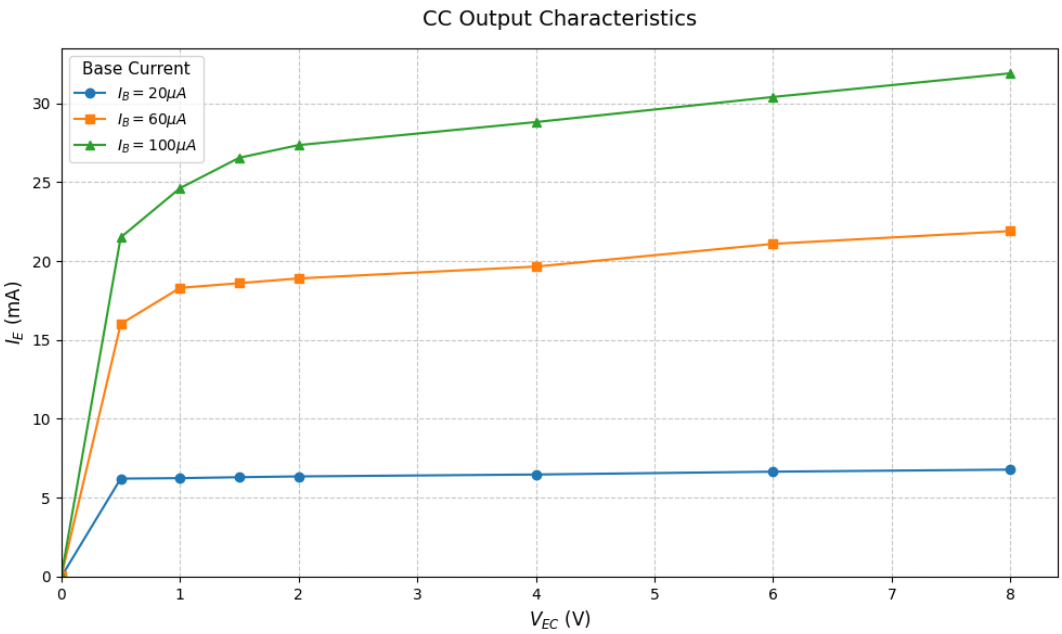



Figure 47: Simulated CC Output Characteristics ($I_E$ vs $V_{EC}$)

## 5.5 Calculations from Observation Data

**Current gain $\gamma$:** Using Table 6 (CC Output Characteristics) at $V_{EC} = 4.0$ V:

$$\gamma_{20\to60\mu A} = \frac{\Delta I_E}{\Delta I_B} = \frac{(19.65 - 6.47)\text{ mA}}{40\ \mu A} \approx 330$$

$$\gamma_{60\to100\mu A} = \frac{(28.81 - 19.65)\text{ mA}}{40\ \mu A} \approx 229$$

$$\gamma_{20\to100\mu A} = \frac{(28.81 - 6.47)\text{ mA}}{80\ \mu A} \approx 279$$

This range (**229–330**) agrees well with theory: using the CE-measured $\beta \approx 342$, $\gamma = \beta + 1 \approx 343$; using the CB-derived $\beta \approx 266$, $\gamma \approx 267$. The measured $\gamma$ range brackets both predictions closely, confirming $\gamma = \beta + 1$ within experimental scatter.

**Output resistance $r_o$:** Using Table 6, between $V_{EC} = 4.0$ V and 8.0 V:

- At $I_B = 20\ \mu A$: $r_o = \dfrac{4.0\text{ V}}{(6.78 - 6.47)\text{ mA}} \approx 12.9\text{ k}\Omega$
- At $I_B = 60\ \mu A$: $r_o = \dfrac{4.0\text{ V}}{(21.9 - 19.65)\text{ mA}} \approx 1.78\text{ k}\Omega$
- At $I_B = 100\ \mu A$: $r_o = \dfrac{4.0\text{ V}}{(31.9 - 28.81)\text{ mA}} \approx 1.29\text{ k}\Omega$

These are the same order of magnitude as the CE-measured $r_o$ (0.66–9.4 k$\Omega$), as expected since $I_E \approx I_C$ makes the CC output curve resemble the CE output curve.

**Input resistance $r_i$:** Using Table 7 (CC Input Characteristics), taking the steepest reliable slope near the conducting region for each $V_{EC}$:

$$r_{i,\,V_{EC}=1V} = \frac{\Delta V_{BC}}{\Delta I_B} = \frac{0.382 - 0.373\text{ V}}{(3.9 - 2.6)\ \mu A} \approx 6.9\text{ k}\Omega$$

$$r_{i,\,V_{EC}=5V} = \frac{4.34 - 4.32 \text{ V}}{(23 - 9.7)\ \mu A} \approx 1.5 \text{ k}\Omega$$

$$r_{i,\,V_{EC}=10V} = \frac{9.38 - 9.36 \text{ V}}{(5.0 - 2.0)\ \mu A} \approx 6.7 \text{ k}\Omega$$

**Note:** this $r_i$ (roughly 1.5–6.9 kΩ) is lower than the CE-measured $r_i$ (13–20 kΩ), which does not match the textbook-expected ordering $r_{i,CB} < r_{i,CE} < r_{i,CC}$, since CC is expected to have the *highest* input resistance of the three. A likely cause is that $V_{BC}$ in Table 7 changes by only very small amounts (a few mV) between consecutive rows while $I_B$ jumps by several $\mu A$, so $r_i = \Delta V_{BC}/\Delta I_B$ is computed over a very short, noise-sensitive voltage interval — small errors in reading $V_{BC}$ to three decimal places get amplified into a large swing in the computed $r_i$. This table should ideally be re-taken with a coarser, more evenly spaced $V_{BC}$ sweep (steps of 0.5 V, as in the original procedure) so that $r_i$ can be computed over a wider, less noise-sensitive interval.

# 6 Results and Conclusion

The input and output characteristics of the BC547 BJT were obtained experimentally, and cross-verified against LTSpice simulation, for the CE, CB, and CC configurations.

From the CE data, $\beta$ was measured in the range 320–363 ($\approx 342$ average), $r_o$ decreased from $\approx 9.4$ kΩ at low $I_C$ to $\approx 0.66$ kΩ at high $I_C$ (consistent with the Early effect), and $r_i$ was found to be $\approx$ 13–20 kΩ near cut-in.

From the CB data, $\alpha = 0.99625$ was measured directly, giving a cross-checked $\beta = \alpha/(1-\alpha) \approx 266$ — in reasonable agreement with the CE-measured $\beta$, given the high sensitivity of that formula to small errors in $\alpha$ near 1. The CB output resistance was $r_o \approx$ 400–800 kΩ, by far the highest of the three configurations, and the CB input resistance was $r_i \approx 830\ \Omega$, the lowest of the three — both exactly as CB theory predicts.

From the CC data, $\gamma$ was measured in the range 229–330, which agrees well with the theoretical prediction $\gamma = \beta + 1$ (giving $\approx 267$ from the CB-derived $\beta$ and $\approx 343$ from the CE-measured $\beta$). The CC output resistance ($\approx$ 1.3–12.9 kΩ) was of similar order to the CE output resistance, as expected since $I_E \approx I_C$. Overall, the relative magnitudes of $r_i$ and $r_o$ across CE and CB matched theory exactly ($r_{i,CB} < r_{i,CE}$ and $r_{o,CB} \gg r_{o,CE} \approx r_{o,CC}$), the $\beta/\alpha$ cross-check agreed within a reasonable margin, and the $\gamma = \beta + 1$ relationship was confirmed by the CC output data.